

DRY-RUN GUIDE | 10_interleave_array.cpp
Source: cpp\2nd_belt\10_interleave_array.cpp

CORE BEHAVIOR

Reads 2N values followed by N, then pairs the first half with the corresponding second-half value.

REPRESENTATIVE INPUT / CALL

Input: size=6; values '1 3 5 2 4 6'; N=3.

STEP TRACE

i=0 prints nums[0],nums[3] => 1,2; i=1 => 3,4; i=2 => 5,6.

EXPECTED OUTPUT / RETURN

'1 2 3 4 5 6'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Given the array nums consisting of 2N elements in the form
004 | [x1, x2, ..., xn, y1, y2, ..., yn], return the array in the form
005 | [x1, y1, x2, y2, ..., xn, yn].
006 |
007 | Input Format:
008 | The first line of input contains the size of an array 2N.
009 | The second line has the elements of the array.
010 | The third line has the value of N.
011 |
012 | Output Format:
013 | The output consists of the array in the form
014 | [x1, y1, x2, y2, ..., xn, yn].
015 | */
016 |
017 | #include <iostream>
018 | #include <vector>
019 | using namespace std;
020 |
021 | int main() {
022 |     int size;
023 |     cin >> size;
024 |
025 |     vector<int> nums(size);
026 |     for (int i = 0; i < size; i++) {
027 |         cin >> nums[i];
028 |     }
029 |
030 |     int n;
031 |     cin >> n;
032 |
033 |     for (int i = 0; i < n; i++) {
034 |         cout << nums[i] << " " << nums[i + n];
035 |         if (i != n - 1) {
036 |             cout << " ";
037 |         }
```

```
038 |     }  
039 |  
040 |     return 0;  
041 | }  
042 |
```